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I Preliminary results

I.1 Monotone operators

An **OPERATOR** is a (possibly multivalued) mapping from an Hilbert space $\mathcal{H}$ to itself, namely $T:\mathcal{H} \rightrightarrows \mathcal{H}$. It can be also thought simply as a subset $T \subseteq \mathcal{H} \times \mathcal{H}$, where the image of x is the set $Tx = \{y \in H : (x,y) \in T\}$. To streamline the notation, Tx can either denote the set image of point x or, as in the following definition, any of its points.

DEFINITION I.1. *An operator $T:\mathcal{H} \to \mathcal{H}$ is [α-**STRONGLY**] **MONOTONE** if $\langle Tx - Ty, x - y\rangle \geq 0$ (resp. $\geq \alpha\|x-y\|^2$) for any $x,y \in \mathcal{H}$. A monotone operator T is **MAXIMAL** if it cannot be extended to a larger monotone operator.*[1]

· *Example* I.2 ([1, Cor. 20.25]). If T is monotone and continuous, then it is maximal monotone.

LEMMA I.3 ([1, Prop. 20.2]). *$T:\mathcal{H} \rightrightarrows \mathcal{H}$ is monotone iff any of the following equivalent properties holds:*

(*a*) $\|x-y+\alpha(Tx-Ty)\| \quad \forall x,y \in \mathcal{H},\ \alpha \in [0,1]$

(*b*) $\|y-Tx\|^2 + \|x-Ty\|^2 \geq \|x-Tx\|^2 + \|y-Ty\|^2$

THEOREM I.4 (Minty, [1, Thm. 21.1]). *A monotone operator T is maximal iff $(\mathbf{I}+T)$ has full rank.*

We define the **INVERSE** of T as the operator $T^{-1} := \{(y,x) : (x,y) \in T\}$. Inversion preserves monotonicity and maximality (see [4, Prop. 3.1]).

LEMMA I.5 ([4, Prop. 3.2,3]). *If T is maximal monotone, then it is a closed subset of $\mathcal{H} \times \mathcal{H}$.*[2] *Besides, it is closed and convex-valued.*

· *Example* I.6 ([4, Thm. 3.2,4]). The following operators are maximal monotone:

- the **subgradient** $\partial f : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ of a convex proper lsc function $f:\mathbb{R}^n \to \overline{\mathbb{R}}$ (it is α-*strongly* monotone iff f is α-strongly convex)
- the **normal cone** $N_C : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ of a nonempty closed convex set $C \subseteq \mathbb{R}^n$

I.2 Firmly nonexpansive operators

We say an operator is **NONEXPANSIVE** if it is 1-Lipschitz-countinuous, that is, if $\|\omega - w\| \leq \|x-y\|$ for every $x,y \in \mathcal{H}$, $\omega \in Tx$ and $w \in Ty$. Notice that any Lipschitz-continuous operator is single-valued, therefore we may use the classical definition of Lipschitz continuity.

DEFINITION I.7. *Given $\alpha \in [0,1]$ we say that $T:\mathcal{H} \to \mathcal{H}$ is α-**AVERAGED** if there exists a nonexpansive operator R s.t. $T = (1-\alpha)\mathbf{I} + \alpha R$.*

REMARK I.8. If $0 \leq \alpha \leq \beta \leq 1$, then α-averagedness is stronger than β-averagedness, that is, if T is α-averaged then it is also β-averaged.

[1] Viewed as a subset $T \subseteq \mathcal{H} \times \mathcal{H}$, this means that any extension $T' \supset T$ is not monotone.

[2] This means that if $y^n \in Tx^n$ for all k, $x^n \to x$ and $y^n \to y$, then $y \in Tx$.

DEFINITION I.9 ([1, Def. 4.1, Prop. 4.2, Cor. 4.3, Rem 4.24]). *We say* $T:\mathcal{H}\to\mathcal{H}$ *is* **FIRMLY NONEXPANSIVE** *if any of the following equivalent properties holds:*[3]

(i) $\|Tx-Ty\|^2 \leq \langle Tx-Ty,\, x-y\rangle$
(ii) $\|Tx\|^2 \leq \langle Tx,\, x\rangle$
(iii) $\|Tx-Ty\|^2 + \|(\mathbf{I}-T)x-(\mathbf{I}-T)y\|^2 \leq \|x-y\|^2$
(iv) $\langle Tx-Ty,\, (\mathbf{I}-T)x-(\mathbf{I}-T)y\rangle \geq 0$
(v) $\|Tx-Ty\| \leq \|\alpha(x-y)+(1-\alpha)(Tx-Ty)\|$ $\forall\alpha\in[0,1]$
(vi) $2T-\mathbf{I}$ is nonexpansive
(vii) T is $1/2$-averaged

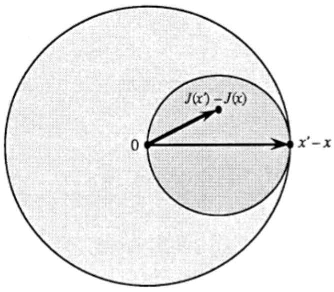

Figure I.1: If J is nonexpansive, then the difference $J(x)-J(x')$ must lie in the larger sphere of radius 1 centered in 0. Firmly nonexpansiveness adds the requirement of lying in the smaller sphere of radius $1/2$ centered in $\frac{x-x'}{2}$.

From [4, Fig.5]

LEMMA I.10 ([1, Prop. 4.2, Cor. 4.3, Ex. 20.5]). *If* T *is firmly nonexpansive, then*

1. $\mathbf{I}-T$ *and* $T^\star$ *are firmly nonexpansive*
2. *the sets of* **FIXED POINTS** $\operatorname{Fix}T$ *and of* **ZEROS** $\operatorname{Zer}T$ *are closed and convex*
3. T *is monotone*

· *Example* I.11 (**Projections**, [1, Prop. 4.11]). Let $C\subseteq\mathcal{H}$ be a closed and convex set. Then

$$\|\Pi_C x-\Pi_C y\|^2 \quad=\quad \langle \Pi_C x-\Pi_C y,\, x-y\rangle$$

and in particular the projection on C is firmly nonexpansive.

DEFINITION I.12. *We say* $T:\mathcal{H}\to\mathcal{H}$ *is* β**-COCOERCIVE** *if* βT *is firmly nonexpansive, that is, if*

$$\beta\|Tx-Ty\|^2 \leq \langle Tx-Ty,\, x-y\rangle$$

LEMMA I.13. *Let* S *be a firmly nonexpansive operator, then for every point* x *we have*

$$\|x-S(x)\|^2 \quad\leq\quad \|x-\zeta\|^2 \;-\; \|S(x)-\zeta\|^2 \qquad \forall\zeta\in\operatorname{Fix}S$$

and, in particular,

$$\|x-\mathsf{S}(x)\|^2 \quad\leq\quad \operatorname{dist}(x,\operatorname{Fix}\mathsf{S})^2 \;-\; \operatorname{dist}(S(x),\operatorname{Fix}\mathsf{S})^2$$

Proof. For all $\zeta\in\operatorname{Fix}S$ we have

$$\begin{aligned}
\|x-\zeta\|^2 \quad&=\quad \|x-S(x)\|^2 \;+\; \|S(x)-\zeta\|^2 \;+\; 2\big\langle x-S(x),\, S(x)-\zeta\big\rangle\\
&=\quad \|x-S(x)\|^2 \;+\; \|S(x)-\zeta\|^2 \;+\; \underbrace{2\big\langle x-\zeta,\, S(x)-\zeta\big\rangle \;-\; 2\big\|S(x)-\zeta\big\|^2}_{\geq 0 \quad \text{(Def. I.9(i))}}\\
&\geq\quad \|x-S(x)\|^2 \;+\; \|S(x)-\zeta\|^2
\end{aligned}$$

By Lemma I.10-2 $\operatorname{Fix}S$ is closed and convex, hence we may choose $\zeta=\zeta^\star:=\Pi_{\operatorname{Fix}S}x$, hence

$$\operatorname{dist}\big(x,\operatorname{Fix}S\big)^2 \quad\geq\quad \|x-S(x)\|^2 \;+\; \|S(x)-\zeta^\star\|^2 \quad\geq\quad \|x-S(x)\|^2 \;+\; \operatorname{dist}\big(S(x),\operatorname{Fix}S\big)^2 \qquad \square$$

[3]It is straightforward to observe that strict nonexpansiveness implies nonexpansiveness, hence the single-valued operator notation.

REMARK I.14. Figure I.2 shows a geometrical interpretation of the inequality

$$\langle x - S(x), \zeta - S(x) \rangle \quad \leq \quad 0 \qquad \forall \zeta \in \mathrm{Fix}\, S \tag{I.14α}$$

which was encountered in the proof of Lemma I.13. We may also infer that if $x \notin \mathrm{Fix}\, S$ and $S(x) \in \mathrm{Fix}\, S$, then $S(x) \in \mathrm{Int}(\mathrm{Fix}\, S)$. Indeed, if $\mathrm{B}(S(x), \varepsilon) \subseteq \mathrm{Fix}\, S$ for some $\varepsilon > 0$, then for $\zeta = \frac{\varepsilon}{2} \frac{S(x)-x}{\|S(x)-x\|} \in \mathrm{Fix}\, S$ the inequality would not hold.

In particular, it follows that *any nontrivial fixed point iteration of a firmly nonexpansive operator S converges to a point $\zeta^\star \in \partial \mathrm{Fix}\, S$.*

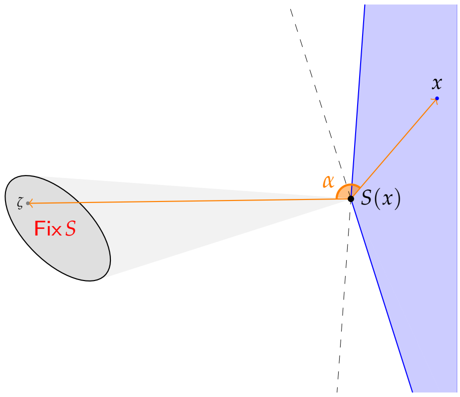

Figure I.2: Displacement of the generating point x given $S(x)$. As shown in the proof, for every $\zeta \in \mathrm{Fix}\, S$ we have $\langle x - S(x), \zeta - S(x) \rangle \leq 0$. It follows that the vectors $x - S(x)$ and $\zeta - S(x)$ cross with an angle $\alpha \geq \pi/2$, and hence x lies in the blue shaded area shown in the picture. In fact, the area is the cone $\{y : \langle y - S(x), \zeta - S(x) \rangle \leq 0 \quad \forall \zeta \in \mathrm{Fix}\, S\}$. Another interpretation is that the blue shaded cone represents the sets of all the points x that for any $\zeta \in \mathrm{Fix}\, S$ satisfy the inequality $\|x - \zeta\| \geq \|S(x) - \zeta\|$.
Together with Figure I.1, this also gives an idea of how $S(x)$ approaches $\mathrm{Fix}\, S$ wrt x.

COROLLARY I.15. *A firmly nonexpansive operator T is* **STRICTLY QUASI-NONEXPANSIVE** *(see §I.3). That is,*

$$\|T(x) - \zeta\| \quad < \quad \|x - \zeta\| \qquad \forall \zeta \in \mathrm{Fix}\, T,\ x \notin \mathrm{Fix}\, T$$

I.3 Fejér monotonicity

A sequence $(x^n)_{n \in \mathbb{N}} \subset \mathcal{H}$ is **FEJÉR MONOTONE** with respect to $C \subseteq \mathcal{H}$ if at each iteration it gets closer to all the points of C, namely, if $\|x^{n+1} - x\| \leq \|x^n - x\|$ for all $n \in \mathbb{N}$ and $x \in C$. Similarly, an operator $T : \mathcal{H} \rightrightarrows \mathcal{H}$ is **FEJÉR MONOTONE** with respect to $C \subseteq \mathcal{H}$ if $\|Tx - z\| \leq \|x - z\|$ for all $x \in \mathcal{H}, z \in C$. If $C = \mathrm{Fix}\, T$ then we say that T is **QUASI NONEXPANSIVE**.

LEMMA I.16 ([1, Prop. 5.4]). *Any Fejér monotone sequence is bounded and its distance from the monotonicity set C decreases.*

THEOREM I.17 ([1, Prop. 5.10, Thm. 5.12]). *Let $(x^n)_{n \in \mathbb{N}}$ be a Fejér monotone sequence with respect to C. If $\mathring{C} \neq \emptyset$, then the sequence is strongly convergent. Moreover, if there exists $\kappa \in [0,1)$ s.t. $\mathrm{dist}(x^{n+1}, C) \leq \kappa\, \mathrm{dist}(x^n, C)$ for all n, then the sequence converges linearly*[4] *to a point $x \in C$.*

THEOREM I.18 (Krasnosel'skiĭ-Mann algorithm, [1, Thm. 5.14, Prop. 5.15]). *Let D be a nonempty closed convex subset of $\mathcal{H}$, $\alpha \in [0,1]$ and $T : D \to D$ an α-averaged operator with $\mathrm{Fix}\, T \neq \emptyset$.*[5] *Let $x_0 \in D$ and $\lambda_n \in [0, 1/\alpha]$ be such that $\sum_{n \in \mathbb{N}} \lambda_n (1 - \alpha \lambda_n) = \infty$. Define the* **KM SEQUENCE** *as*

$$x^{n+1} \quad = \quad \left[\lambda_n \mathbf{I} + (1 - \lambda_n) T\right] x^n$$

[4] More precisely, $\|x^n - x\| \leq 2\kappa^n \mathrm{dist}(x^0, C)$.
[5] $\alpha = 1$ corresponds to T simply being nonexpansive.

Then,

1. (x^n) *is Fejér monotone with respect to* $\mathsf{Fix}\,T$
2. $Tx^n - x^n$ *converges (strongly) to* 0
3. x^n *converges weakly to a point* $x \in \mathsf{Fix}\,T$

· *Example* I.19 ([1, Ex. 5.17]). For $\alpha = 1/2$ we may infer that the fixed point iterations $x^{n+1} = Tx^n$ of a firmly nonexpansive operator T converge weakly to a point in $\mathsf{Fix}\,T$.

I.4 Metric subregularity

DEFINITION I.20. *An operator* $T : \mathcal{H} \rightrightarrows \mathcal{H}$ *is said to be (globally)* **METRIC SUBREGULAR** *(wrt* $\mathsf{Fix}\,S$ *) if there exists* $\kappa > 0$ *s.t.*

$$\mathsf{dist}(x, \mathsf{Fix}\,T) \;\leq\; \kappa \|x - Tx\| \qquad \forall x \in \mathcal{H}$$

Notice that if T is L-Lipschitz continuous and κ-metric subregular, then $\kappa \geq L$. In particular every metric subregular nonexpansive mapping has subregularity coefficient $\kappa \geq 1$.

II Splitting schemes

II.1 Resolvents

Given $\lambda>0$ and $T:\mathcal{H}\rightrightarrows\mathcal{H}$, the operator $\mathcal{J}_{\lambda T} := (\mathbf{I}+\lambda T)^{-1}$ is called a **RESOLVENT** of T.

THEOREM II.1 ([4, Thm. 3.6, Cor. 3.6.2]). *An operator T is (maximal) monotone iff its resolvent $\mathcal{J}_{\lambda T}$ is firmly nonexpansive (with full domain), where λ is any strictly positive scalar. In particular, the resolvents of a monotone operator are single-valued.*

COROLLARY II.2 ([4, Cor. 3.6.3]). *Let $\lambda>0$ and let T be a monotone [resp. maximal monotone] operator on $\mathcal{H}$. Then, each element $z\in\mathcal{H}$ can be written in at most [resp. exactly] one way as $z=x+\lambda y$ with $y\in Tx$.*

The **REFLECTED RESOLVENT** of a monotone operator T is $(2\mathcal{J}_T - \mathbf{I})$. Since $\mathcal{J}_T = \frac{1}{2}(2\mathcal{J}_T-\mathbf{I})+\frac{1}{2}\mathbf{I}$ is firmly nonexpansive, it follows that the reflected resolvent is nonexpansive. Besides, a simple computation shows that the reflected resolvent acts on $\mathcal{H}$ as a *conjugator wrt* T, that is, it "flips" the sign of y in the expression $z=x+y$ with $y\in Tx$. More precisely,

$$z \;=\; x+y \quad \text{with } y\in Tx \quad (\text{i.e. } z=\mathcal{J}_T x) \qquad \Rightarrow \quad (2\mathcal{J}_T-\mathbf{I})z \;=\; x-y \tag{II.3}$$

If T is maximal monotone, then [4, Prop. 3.9]

$$\operatorname{Zer} T \;=\; \operatorname{Fix}\mathcal{J}_{\lambda T} \qquad \forall\lambda>0 \tag{II.4}$$

and its resolvent is related to that of its inverse as [1, Prop. 23.18]

$$\mathcal{J}_{T^{-1}} \;=\; \mathbf{I}-\mathcal{J}_T \tag{II.5}$$

· *Example* II.6 ([4, Prop. 3.11]). $\mathcal{P}_C=\mathcal{J}_{\lambda N_C}$ for every closed and convex set $C\subseteq\mathcal{H}$ and $\lambda>0$.[6]

LEMMA II.7 ([1, Prop. 23.28]). *If $T:\mathcal{H}\to\mathcal{H}$ is maximal monotone, then*

1. $\mathcal{J}_T \;=\; \mathcal{J}_{\lambda T}\circ\big(\lambda\mathbf{I}+(1-\lambda)\mathcal{J}_T\big) \quad \forall\lambda>0$
2. $\|\mathcal{J}_{\lambda T}x-x\|^2 \le (2-\lambda)\|\mathcal{J}_T x-x\|^2 \quad \forall\lambda\in(0,1]$
3. $\|\mathcal{J}_T x-\mathcal{J}_{\lambda T}x\|^2 \le |1-\lambda|\,\|\mathcal{J}_T x-x\|^2 \quad \forall\lambda>0$

II.2 Proximal point iterations

THEOREM II.8 (Proximal Point algorithm [1, Thm. 23.41]). *Let T be a maximal monotone operator with $\operatorname{Zer} T\neq\emptyset$, and let $\gamma_k>0$ be s.t. $\sum_{k\in\mathbb{N}}\gamma_k^2 = +\infty$. Then, for any starting point x^0 the* **PROXIMAL POINT** *iteration*

$$x^{k+1} \;=\; \mathcal{J}_{\gamma_k T}x^k \;=\; \big(I+\gamma_k T\big)^{-1}x^k \tag{PP}$$

converges weakly to a point $x^\star\in\operatorname{Fix} T$.

Proximal point interations are made of *backward steps*, in that they involve the inverse of an operator. It follows that in general each step is more difficult to compute than the corresponding *forward step* $x^{k+1}=(\mathbf{I}-\gamma T)x^k$. On the other hand proximal point algorithms require less restrictive conditions both on the stepsize and the structure of the problem. When computing forward steps, it might also occur to end up in a point x^k s.t. $Tx^k=\emptyset$, which is not the case for the corresponding backward step provided T is maximal monotone.

[6] In fact, for every $\lambda>0$ $N_C=\lambda N_C$.

Theorem II.9 (Inexact Proximal Point algorithm [4, Thm. 3.8]). *Let* T *be a maximal monotone operator and let* $\gamma_k > 0$, $\varepsilon_k \geq 0$ *and* $x^k \in \mathcal{H}$ *be sequences satisfying* $\forall k \in \mathbb{N}$

- $\|x^{k+1} - (\mathbf{I} + \gamma_k T)^{-1} x^k\| \leq \varepsilon_k$
- $\inf \gamma_k > 0$
- $\sum_{k \in \mathbb{N}} \varepsilon_k < \infty$

If $\operatorname{Zer} T \neq \emptyset$, *then for any starting point* x^0 *the* **inexact proximal point** *iteration* x^k *converges weakly to a point* $x^\star \in \operatorname{Fix} T$; *otherwise,* $\|x^k\| \to \infty$.

The following result applies only to finite dimensional spaces; it is indeed evident that $\mathbb{R}^n$ can be replaced by any finite dimensional Hilbert space.

Theorem II.10 (Averaged Inexact Proximal Point algorithm [4, Thm. 3.9]). *Let* $T : \mathbb{R}^n \to \mathbb{R}^n$ *be a maximal monotone operator with* $\operatorname{Zer} T \neq \emptyset$ *and let* $\gamma > 0$. *Let* $\rho_k > 0$, $\varepsilon_k \geq 0$, $x^k, y^k \in \mathcal{H}$ *be sequences satisfying* $\forall k \in \mathbb{N}$

- $\|y^k - (\mathbf{I} + \gamma T)^{-1} x^k\| \leq \varepsilon_k$
- $\sum_{k \in \mathbb{N}} \varepsilon_k < \infty$
- $0 < \inf \rho_k \leq \sup \rho_k < 2$
- $x^{k+1} = (1 - \rho_k) x^k + \rho_k y^k$

Then, for any starting point x^0 *the* **averaged inexact proximal point** *iteration* x^k *converges (strongly) to a point* $x^\star \in \operatorname{Fix} T$.

- *Example* II.11 ([1, Ex. 23.3]). The resolvent of the subgradient of a lsc proper convex function $f : \mathcal{H} \to \overline{\mathbb{R}}$ is the **proximal mapping**

$$\mathcal{J}_{\gamma \partial f}(x) \quad = \quad \operatorname{prox}_{\gamma f}(x) \quad := \quad \operatorname*{argmin}_{y \in \mathcal{H}} \left\{ f(y) \; + \; \tfrac{1}{2\gamma} \|x - y\|^2 \right\} \tag{II.11α}$$

and its fixed point iterations consist in the **proximal point algorithm**.

II.3 Forward-Backward Splitting

Splitting an operator comes in handy when computing the resolvents of an operator T is too challenging. The idea is to *split* T as the sum of two operators $T = A + B$. Though it might not be necessary, working with maximal monotone operators is always beneficial.

Theorem II.12 ([1, Cor. 24.4]). *Let* $A, B : \mathcal{H} \to \mathcal{H}$ *be maximal monotone. Then* $A + B$ *is maximal monotone provided one of the following properties holds:*

[a] $\operatorname{dom} B = \mathcal{H}$
[b] $\operatorname{dom} A \cap \operatorname{dom} B \neq \emptyset$
[c] $0 \in \operatorname{int}(\operatorname{dom} A \setminus \operatorname{dom} B)$
[d] $\forall x \in \mathcal{H} \;\; \exists \varepsilon > 0$ *s.t.* $[0, \varepsilon x] \subseteq \operatorname{dom} A \setminus \operatorname{dom} B$

Forward-Backward Splitting relies on the following result:

Lemma II.13 ([4, Prop. 3.13]). *If* $A, B : \mathcal{H} \to \mathcal{H}$ *are monotone operators and* B *is maximal, then*

$$\operatorname{Fix} (\mathbf{I} + \lambda B)^{-1} (\mathbf{I} - \lambda A) \quad = \quad \operatorname{Zer}(A + B) \qquad \forall \lambda > 0 \tag{II.13α}$$

Forward-Backward iterations are indeed those of the form[7]

$$x^{k+1} \quad \in \quad (\mathbf{I} + \lambda_k B)^{-1} (\mathbf{I} - \lambda_k A) x^k \tag{FBS}$$

[7] Algorithms of this kind include the *gradient projection method*, because of the identity $\mathcal{J}_{N_C} = \mathcal{P}_C$ that holds for every nonempty closed and convex set C, where N_C is its normal cone.

In general, if A is single-valued, then the '$\in$' sign turns into an equality, and hence if $x^j \in \mathsf{Zer}(A+B)$ it follows that $x^{j+1} = x^j$. If A is not single valued, such implication does not hold, in that it may depend on the selected *choice*.

Algorithm I Forward-Backward Splitting scheme with constant stepsizes γ and λ_k-averaged steps.

Choose $x^0 \in \mathcal{H}$
repeat
  $y^k \leftarrow x^k - \gamma B x^k$
  $x^{k+1} \leftarrow x^k + \lambda_k(\mathcal{J}_{\gamma A} y^k - x^k)$
until $x^k \approx x^{k-1}$
return $x^k \underset{\approx}{\in} \mathsf{Fix}(A+B)$

Theorem II.14 ([1, Thm. 25.8]). *Let $A: \mathcal{H} \rightrightarrows \mathcal{H}$ be maximal monotone and $B: \mathcal{H} \to \mathcal{H}$ be β-cocoercive, such that $\mathsf{Zer}(A+B) \neq \emptyset$. Let $\gamma \in (0, 2\beta)$, set $\delta = \frac{1}{2} + \min\{1, \beta/\gamma\}$, and let $(\lambda_n)_{n\in\mathbb{N}} \subset [0,\delta]$ be a sequence s.t. $\sum_n \lambda_2(\delta - \lambda_n) = \infty$. Then, for every starting point $x^0 \in \mathcal{H}$, the sequence $(x^k)_{k\in\mathbb{N}}$ generated by the Forward-Backward Splitting Algorithm I converges weakly to a point $x^\star \in \mathsf{Fix}(A+B)$.*

Theorem II.15 (Inexact FBS. [3, Lem. 4.4]). *Let A, B and $(\lambda_k)_{k\in\mathbb{N}}$ be as in Theorem II.14 with $\delta = 2 - \gamma/2\beta$. Let $(e_A^k)_{k\in\mathbb{N}}, (e_B^k)_{k\in\mathbb{N}} \subset \mathcal{H}$ be such that $\sum_k \lambda_k(\|e_A^k\| + \|e_B^k\|) < \infty$. Then, for any starting point $x^0 \in \mathcal{H}$, the iterations*

$$x^{k+1} = (1-\lambda_k)x^k + \lambda_k\Big(\mathcal{J}_A(x^k - \gamma(Bx^k + e_B^k)) + e_A^k\Big) \tag{II.15α}$$

converge weakly to a point $x^\star \in \mathsf{Zer}(A+B)$.

II.4 Peaceman-Rachford Splitting

This kind of splitting scheme is obtained by interchanging the roles of A and B at each iteration. Other than T, also both the splitting operators A and B are required to be maximal monotone, so that at each iteration the backward steps are always single-valued (and not empty-valued). The Peaceman-Rachford Splitting has the following form:

$$x^{k+1} \in (\mathbf{I}+\lambda_k B)^{-1}(\mathbf{I}-\lambda_k A)(\mathbf{I}+\lambda_k A)^{-1}(\mathbf{I}-\lambda_k B)\, x^k \tag{PRS}$$

If A or B is multivalued, then the inclusion sign '$\in$' is needed, giving rise to a non uniquely determined algorithm. Anyway, there is a slight modification of the algorithm that overcomes the issue: we refer to such algorithm as the *tight* (or *canonical*) PRS, that can be summarized in the following 2 steps [4, p. 77]: starting from any $x^0 \in \mathcal{H}$ and $b^0 \in Bx^0$,

1. find the unique y^{k+1}, a^{k+1} with $a^{k+1} \in Ay^{k+1}$ such that $y^{k+1} + \lambda_k a^{k+1} = x^k - \lambda_k b^k$
2. find the unique x^{k+1}, b^{k+1} with $b^{k+1} \in Bx^{k+1}$ such that $x^{k+1} + \lambda_k b^{k+1} = y^{k+1} - \lambda_k a^{k+1}$

Using (II.3) and choosing a constant stepsize sequence $\lambda_k \equiv \lambda$, we may express (PRS) in terms of the sequence

$$z^{k+1} = \big(2\mathcal{J}_{\lambda A} - \mathbf{I}\big)\big(2\mathcal{J}_{\lambda B} - \mathbf{I}\big) z^k \tag{PRS'}$$

which is related to the (PRS) sequence x^k via $x^k = \mathcal{J}_{\lambda B} z^k$, and in particular [4, Prop. 3.15]

$$z \in \mathsf{Fix}\Big[\big(2\mathcal{J}_{\lambda A} - \mathbf{I}\big)\big(2\mathcal{J}_{\lambda B} - \mathbf{I}\big)\Big] \quad\Leftrightarrow\quad x = \mathcal{J}_{\lambda B}\, z \in \mathsf{Zer}\, T \tag{II.16}$$

Notice that, since computing $\mathcal{J}_{\lambda B} z^k$ is necessary in order to retrieve z^{k+1}, once z^{k+1} has been computed x^k is immediately available too.

Algorithm II Peaceman-Rachford Splitting scheme

Choose $z^0 \in \mathcal{H}$

repeat

$x^k \leftarrow \mathcal{J}_{\lambda B}\, z^k$

$y^k \leftarrow \mathcal{J}_{\lambda A}\,(2x^k - z^k)$

$z^{k+1} \leftarrow z^k + 2(y^k - x^k)$

until $z^{k+1} \approx z^k$

return $x^k \underset{\sim}{\in} \mathsf{Fix}(A+B)$

Theorem II.17 ([4, Cor. 3.17.1]). *Let $A, B : \mathbb{R}^n \to \mathbb{R}^n$ be maximal monotone operators such that $\mathsf{Zer}\,(A+B) \neq \emptyset$. If at least one is strongly monotone and single-valued, then the (PRS′) method (or equivalently, (PRS)), converges.*

II.5 Douglas-Rachford Splitting

The Douglas-Rachford splitting scheme consists of the following steps:

$$x^{k+1} \quad = \quad (\mathbf{I} + \lambda B)^{-1}\Big[(\mathbf{I} + \lambda A)^{-1}(\mathbf{I} - \lambda B) \; + \; \lambda B\Big]x^k \tag{DRS}$$

We may draw the same conclusions of the (PRS) scheme, such as preferring an equivalent formulation which will be given in (DRS′). Besides, then again multivalued mappings cause the sequence not to be unique, but all the convergence result concern the *canonical* version whose 2 steps are listed below (the first step coincides indeed with the (PRS) scheme): starting from any $x^0 \in \mathcal{H}$ and $b^0 \in Bx^0$

1. find the unique y^{k+1}, a^{k+1} with $a^{k+1} \in Ay^{k+1}$ such that $y^{k+1} + \lambda_k a^{k+1} = x^k - \lambda_k b^k$
2. find the unique x^{k+1}, b^{k+1} with $b^{k+1} \in Bx^{k+1}$ such that $x^{k+1} + \lambda_k b^{k+1} = y^{k+1} + \lambda_k b^k$

Then again, if the stepsize is constant, it is also possible to express the (DRS) scheme in terms of the sequence

$$z^{k+1} \quad = \quad \Big[\underbrace{\mathcal{J}_{\lambda A}\,(2\mathcal{J}_{\lambda B} - \mathbf{I}) \; + \; (\mathbf{I} - \mathcal{J}_{\lambda B})}_{:=\ \mathbf{G}_{\lambda,A,B}}\Big]z^k \tag{DRS′}$$

which is related to the (DRS) steps in exactly the same way discussed for the (PRS) scheme [4, Prop. 3.19]:

$$x^k \quad = \quad \mathcal{J}_{\lambda B}\, z^k \tag{II.18a}$$

and in particular

$$z \;\in\; \mathsf{Fix}\,\mathbf{G}_{\lambda,A,B} \quad \Leftrightarrow \quad x \,=\, \mathcal{J}_{\lambda B}\, z \;\in\; \mathsf{Zer}\,T \tag{II.18b}$$

Besides, it is straightforward to observe that the operator $\mathbf{G}_{\lambda,A,B}$ in (PRS′) is the $\frac{1}{2}$-average of the nonexpansive (PRS′) operator, and hence it is firmly nonexpansive:

$$\mathbf{G}_{\lambda,A,B} \quad = \quad \frac{1}{2}\Big[\mathbf{I} \; + \; (2\mathcal{J}_{\lambda A} - \mathbf{I})\,(2\mathcal{J}_{\lambda B} - \mathbf{I})\Big] \tag{II.19}$$

This marks the main difference between the (PRS) method, which converges only under strong assumptions, and the (DRS) method which is indeed among the most widely used splitting schemes. The iterations of the (DRS) algorithm can be summarized as follows:

Theorem II.20 ([1, Thm. 25.6]). *Let $A, B : \mathbb{R}^n \to \mathbb{R}^n$ be maximal monotone operators such that $\mathsf{Zer}(A+B) \neq \emptyset$. Let $\gamma > 0$, $x_0 \in \mathcal{H}$, and $(\lambda_k)_{k\in\mathbb{N}} \subset [0,2]$ such that $\sum_{k\in\mathbb{N}} \lambda_k(2-\lambda_k) = \infty$. Let x^k, y^k and z^k be defined by the Douglas-Rachford Splitting Algorithm III. Then, $z^k - x^k$ converges strongly to 0 and x^k converges weakly to a point $x^\star \in \mathsf{Fix}(A+B)$.*

Algorithm III Douglas-Rachford Splitting scheme (with non constant stepsizes λ)

Choose $z^0 \in \mathcal{H}$ and $(\lambda_k)_{k\in\mathbb{N}}$
repeat
$x^k \leftarrow \mathcal{J}_{\lambda B}\, z^k$
$y^k \leftarrow \mathcal{J}_{\lambda A}(2x^k - z^k)$
$z^{k+1} \leftarrow y^k + \lambda_k(z^k - x^k)$
until $z^{k+1} \approx z^k$
return $x^k \underset{\sim}{\in} \mathsf{Fix}(A+B)$

II.6 Primal-Dual Splitting

We now focus on optimization problems of the form

$$\underset{x\in\mathbb{R}^n}{\mathsf{minimize}}\, f(x) + g(\mathbf{L}x) \tag{P}$$

where $f:\mathbb{R}^n \to \overline{\mathbb{R}}$ and $f:\mathbb{R}^m \to \overline{\mathbb{R}}$ are lsc proper convex functions and $\mathbf{L}:\mathbb{R}^n \to \mathbb{R}^m$ is linear. We may as well think of $\mathbf{L}$ as a matrix $\mathbf{L}\in\mathbb{R}^{n\times m}$, and assume that g embeds all the constraints of the problem (via, for instance, indicator functions). Problem (P) is equivalent to finding zero of the operator $T = \partial(f + g\circ\mathbf{L}) \supseteq \partial f + \partial g \circ \mathbf{L}$. The following result is a consequence of such inclusion.

Lemma II.21 ([4, Prop. 3.21]). *Any $x^\star \in \mathsf{Zer}(\partial f + \partial g\circ\mathbf{L})$ is an optimal solution to (P). Moreover, if any of the following conditions*

[*a*] $\mathsf{ri}(\mathsf{dom}\, f) \cap \mathsf{ri}(\mathbf{L}^{-1}\mathsf{dom}\, g) \neq \emptyset$ [8]
[*b*] *f is* **PSEUDOPOLYHEDRAL**[9] *and* $\mathsf{dom}\, f \cap \mathsf{ri}(\mathbf{L}^{-1}\mathsf{dom}\, g) \neq \emptyset$
[*c*] *g is pseudopolyhedral and* $\mathsf{ri}(\mathsf{dom}\, f) \cap \mathbf{L}^{-1}\mathsf{dom}\, g \neq \emptyset$
[*d*] *f and g are both pseudopolyhedral and problem (P) is feasible*

is satisfied, then also the converse holds, that is, any optimal solution of (P) is a zero of $\partial f + \partial g\circ\mathbf{L}$.

Problem (P) can be also written in a constrained form as $\underset{x\in\mathbb{R}^n}{\mathsf{minimize}}\ f(x) + g(z)$ s.t. $z = \mathbf{L}x$, whose dual formulation in terms of the convex conjugates of f and g is [4, p. 97]

$$\underset{y\in\mathbb{R}^m}{\mathsf{minimize}}\, f^\star(-\mathbf{L}^\mathsf{T} y) + g^\star(y) \tag{P*}$$

Let us recall that the **CONVEX CONJUGATE** of a proper convex lsc function $f:\mathbb{R}^n \to \mathbb{R}$ is the convex function

$$\begin{aligned} f^\star : \mathbb{R}^n &\longrightarrow \overline{\mathbb{R}} \\ y &\longmapsto \sup_{x\in\mathbb{R}^n} \{\langle x, y\rangle - f(x)\} \end{aligned} \tag{II.22}$$

which is characterized as follows:

Theorem II.23 (Conjugate Subgradient Theorem. [4, Thm. 3.16]). *Let $f:\mathbb{R}^n \to \mathbb{R}$ be a lsc proper convex function. Then, the following statements are equivalent:*

(*i*) $f(x) + f^\star(y) = \langle x, y\rangle$
(*ii*) $y \in \partial f(x)$
(*iii*) $x \in \partial f^\star(y)$

Theorem II.24 (Moreau's identity. [1, Thm. 14.3]). *Let $f:\mathbb{R}^n \to \overline{\mathbb{R}}$ be a lsc proper convex function. Then, for all $\gamma > 0$*

$$\mathsf{prox}_{\gamma f^\star}(x) = x - \gamma\mathsf{prox}_{f/\gamma}(x/\gamma) \tag{II.24α}$$

[8] $\mathbf{L}^{-1}$ denotes the inverse in the operator sense. In matrix notation, given a set $E \subseteq \mathbb{R}^m$ $\mathbf{L}^{-1}E := \{x\in\mathbb{R}^n : \mathbf{L}x \in E\}$.
[9] f is pseudopolyhedral if it is convex and its proper domain is polyhedral

Theorem II.25 (Karush-Kuhn-Tucker. [4, Prop. 3.26]). *The following conditions are equivalent:*

(*i*) $x^\star \in \mathbb{R}^n$ *is optimal for* (P) *and* $y^\star \in \mathbb{R}^n$ *is optimal for* (P*)

(*ii*) $x^\star \in \mathbb{R}^n$ *is optimal for* (P), $y^\star \in \partial g^\star(\mathbf{L}x^\star)$ *and* $-\mathbf{L}^\mathsf{T} y^\star \in \partial f^\star(x^\star)$

(*iii*) $y^\star \in \mathbb{R}^m$ *is optimal for* (P*), $x^\star \in \partial f^\star(-\mathbf{L}y^\star)$ *and* $\mathbf{L}x^\star \in \partial g^\star(y^\star)$

(*iv*) $-\mathbf{L}^\mathsf{T} y^\star \in \partial f(x^\star)$ *and* $y^\star \in \partial g(\mathbf{L}x^\star)$

(*v*) $f(x^\star) + g(\mathbf{L}x^\star) = -\left(f^\star(-\mathbf{L}^\mathsf{T} y^\star) + g^\star(y^\star)\right)$

In light of the Conjugate Subgradient Theorem II.23, the KKT condition II.25(iv) is equivalent to $-\mathbf{L}^\mathsf{T} y^\star \in \partial f(x^\star)$ and $\mathbf{L}x^\star \in \partial g^\star(y^\star)$, that is,

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \in \Bigg[\underbrace{\begin{pmatrix} & -\mathbf{L}^\mathsf{T} \\ \mathbf{L} & \end{pmatrix}}_{A} + \underbrace{\begin{pmatrix} \partial f \\ & \partial g^\star \end{pmatrix}}_{B} \Bigg] \begin{pmatrix} x^\star \\ y^\star \end{pmatrix} \tag{II.26}$$

Splittings on the operator above are called **primal-dual**, in that they act on primal and dual variables x and y at the same time. Let us observe that in light of Example II.11, the resolvent $\mathcal{J}_B$ of B separates into two proximal mappings:

$$\mathcal{J}_{\gamma B} = \left(\mathrm{prox}_{\gamma f}, \mathrm{prox}_{\gamma g^\star}\right) \tag{II.27a}$$

while that of A is simply

$$\mathcal{J}_{\gamma A} = \begin{pmatrix} \gamma \mathbf{I}_n & -\mathbf{L}^\mathsf{T} \\ \mathbf{L} & \gamma \mathbf{I}_m \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{Q}_\gamma & -\gamma \mathbf{L}^\mathsf{T} \mathbf{R}_\gamma \\ \gamma \mathbf{L} \mathbf{Q}_\gamma & \mathbf{R}_\gamma \end{pmatrix} \quad \text{where} \quad \begin{aligned} \mathbf{Q}_\gamma &= \left[\mathbf{I}_n + \gamma^2 \mathbf{L}^\mathsf{T} \mathbf{L}\right]^{-1} \\ \mathbf{R}_\gamma &= \left[\mathbf{I}_m + \gamma^2 \mathbf{L} \mathbf{L}^\mathsf{T}\right]^{-1} \end{aligned} \tag{II.27b}$$

Algorithm IV Primal-Dual Forward-Backward Splitting corresponding to the splitting given in (II.26).

Choose $x^0 \in \mathbb{R}^n$, $y^0 \in \mathbb{R}^m$ and $(\gamma_k)_{k\in\mathbb{N}}$
repeat
$x^{k+1} \leftarrow \mathrm{prox}_{\gamma_k f}(x^k - \gamma_k \mathbf{L}^\mathsf{T} y^k)$
$y^{k+1} \leftarrow \mathrm{prox}_{\gamma_k g^\star}(y^k + \gamma_k \mathbf{L}^\mathsf{T} x^k)$
until $(x^k, y^k) \approx (x^{k-1}, y^{k-1})$
return (x^k, y^k) $\sim$ primal dual optimal pair of (P)-(P*)

As to the Douglas-Rachford Splitting scheme, differently from the general formulation given in Algorithm III, variable x is now replaced by the couple (x, y) (notice that y now has a different role), and variable z by (z, w). Using (II.27b), after some algebra we may omit assigning what in Algorithm III was variable y, and summarize the Primal-Dual DRS algorithm as follows.

Algorithm V Primal-Dual Douglas-Rachford Splitting corresponding to the splitting given in (II.26).

Choose $x^0 \in \mathbb{R}^n$, $y^0 \in \mathbb{R}^m$ and $(\gamma_k)_{k\in\mathbb{N}}$
repeat
$\mathbf{Q} \leftarrow \left[\mathbf{I}_n + \gamma_k^2 \mathbf{L}^\mathsf{T} \mathbf{L}\right]^{-1}$ % (if γ_k is not constant)
$\mathbf{R} \leftarrow \left[\mathbf{I}_m + \gamma_k^2 \mathbf{L} \mathbf{L}^\mathsf{T}\right]^{-1}$ % (if γ_k is not constant)
$x^k \leftarrow \mathrm{prox}_{\gamma_k f}(z^k)$
$y^k \leftarrow \mathrm{prox}_{\gamma_k g^\star}(w^k)$
$z^{k+1} \leftarrow \gamma_k(z^k - x^k) + \mathbf{Q}(2x^k - z^k) - \gamma_k \mathbf{L}^\mathsf{T} \mathbf{R} \gamma_k (2y^k - w^k)$
$w^{k+1} \leftarrow \gamma_k(w^k - y^k) + \gamma_k \mathbf{L} \mathbf{Q} \gamma_k (2x^k - z^k) + \mathbf{R}(2y^k - w^k)$
until $(z^{k+1}, w^{k+1}) \approx (z^k, w^k)$
return (x^k, y^k) $\sim$ primal-dual optimal pair of (P)-(P*)

II.6.1 Adding a smooth term

We will now generalize the class of optimization problems by means of adding a smooth term to the cost function. Namely, we consider (feasible) problems of the form

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \left\{ F(x) + g(x) + h(\mathbf{L}x) \right\} \tag{P3}$$

where

- $F:\mathbb{R}^n \to \mathbb{R}$ is convex, differentiable, and with β-Lipschitz countinuous gradient;
- $g:\mathbb{R}^n \to \mathbb{R}$ and $h:\mathbb{R}^m \to \mathbb{R}$ are proximable lsc proper convex functions;
- $\mathbf{L}\in\mathbb{R}^{m\times n}$ is a linear operator.

The case $F\equiv 0$ corresponds to problem (P). The dual is

$$\underset{y\in\mathbb{R}^m}{\text{minimize}} \left\{ (F+g)^\star(-\mathbf{L}^\mathsf{T}y) + h^\star(y) \right\} \tag{P3*}$$

This formulation has the main disatvantage of involving the proximal mapping of $(F+g)$, which is not easy to compute even assuming that of F and g are. Condat in [3] created a splitting scheme that tackles the problem by applying only ∇F and the proximal mappings of g and h, without any other implicit operator or inner loop. In particular, no inverse operator of the form $(1+\gamma_k\mathbf{L}\mathbf{L}^\mathsf{T})^{-1}$ is required, as was the case of the Douglas-Rachford Algorithm V.

Algorithm VI Condat's inexact λ_k-averaged primal-dual splitting scheme for problem (P3).

Choose suitable stepsizes σ,τ satisfying Thm. II.28's hp
Choose a starting pair $(x^0,y^0)\in\mathbb{R}^n\times\mathbb{R}^m$
repeat
$\quad \tilde{x}^{k+1} \leftarrow \operatorname{prox}_{\tau g}\big(x^k - \tau\big(\nabla F(x^k)+e_F^k + \mathbf{L}^\mathsf{T}y^k\big)\big) + e_g^k$
$\quad \tilde{y}^{k+1} \leftarrow \operatorname{prox}_{\sigma h^\star}\big(y^k + \sigma\mathbf{L}(2\tilde{x}^{k+1}-x^k)\big) + e_h^k$
$\quad x^{k+1} \leftarrow (1-\lambda_k)x^k + \lambda_k\tilde{x}^{k+1}$
$\quad y^{k+1} \leftarrow (1-\lambda_k)y^k + \lambda_k\tilde{y}^{k+1}$
until $(x^k,y^k) \approx (x^{k-1},y^{k-1})$
return $(x^k,y^k) \sim$ primal-dual solution of (P3)-(P3*)

Theorem II.28 ([3, Thm. 3.1]). *Let $\sigma,\tau>0$ and the sequences $(\lambda_k)_{k\in\mathbb{N}}\subset\mathbb{R}$, $(e_k^F)_{k\in\mathbb{N}}$, $(e_k^g)_{k\in\mathbb{N}}\subset\mathbb{R}^n$ and $(e_k^h)_{k\in\mathbb{N}}\subset\mathbb{R}^m$ be the parameters of Algorithm VI, and let $\beta>0$ be the Lipschitz constant of ∇F. Suppose that the following hold:*

- $\frac{1}{\tau}-\sigma\|\mathbf{L}\|^2 \geq \frac{\beta}{2}$
- $\lambda_k\in(0,\delta)$ *where* $\delta := 2-\frac{\beta}{2}\left(\frac{1}{\tau}-\sigma\|\mathbf{L}\|^2\right)^{-1} \in [1,2]$
- $\sum_k \lambda_k(\delta-\lambda_k) = \infty$
- $\sum_k \left(\|e_F^k\|+\|e_g^k\|+\|e_h^k\|\right)\lambda_k < \infty$

Then, the sequence (x^k,y^k) generated by Algorithm VI converges to a primal-dual pair $(x^\star,y^\star)$ solution of (P3)-(P3*).